Enabling an Open Near-Package Ecosystem for AI Scale-Up

Bill Wilson

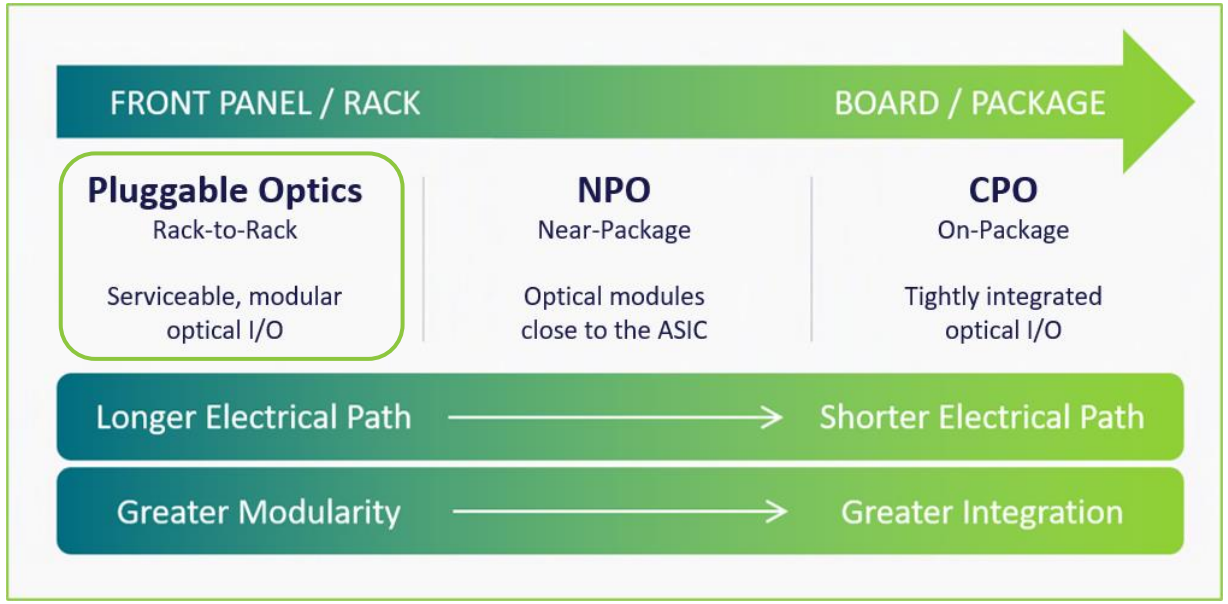
Group Manager, New Product Development

molex



The Copper-Optics Transition Is Moving Inside the Box

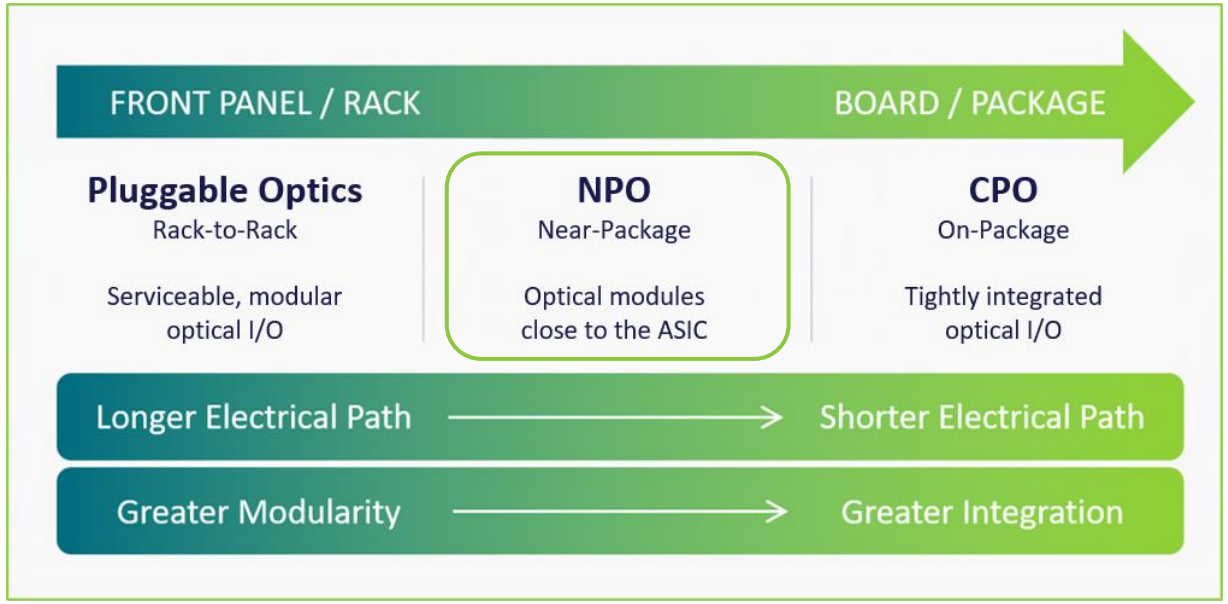
- Historically, copper-versus-optics decisions were made at the front panel, based primarily on reach, cost, power, and performance.
- AI scale-out has accelerated adoption of pluggable optics, while passive and active copper remain important for shorter links.
- Yet, as accelerator fabrics grow in size, bandwidth, and complexity, the copper-optics boundary is exploring a shift from the front panel to inside the box.



Takeaway
Diverse architecture pathways are emerging at progressively shorter reaches in areas of the system channel that have historically relied on copper-centric solutions.

NPO is Gaining Momentum

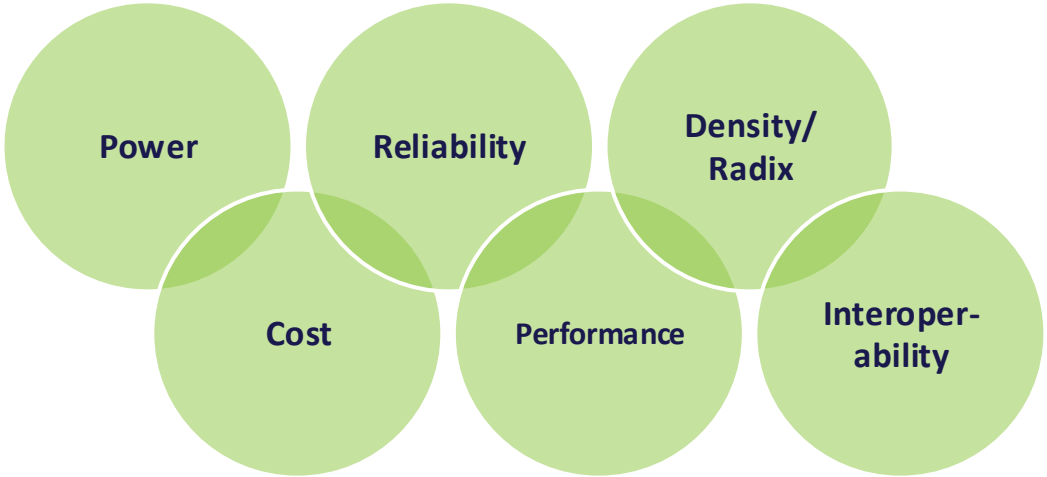
- Near-Package Optics (NPO) is emerging as a practical balance between traditional pluggable optics and fully integrated optical architectures.
- Moving the optical engine closer to the ASIC shortens the high-speed electrical channel while retaining a modular, serviceable interface in a lower risk area than on-package.
- System designers are balancing multiple criteria as they develop 224G and 448G architectures:
  - Power
  - Performance / latency
  - Density / radix
  - Serviceability
  - Architectural Flexibility



Takeaway
NPO combines shorter electrical reach and improved power efficiency with the modularity, serviceability, and architectural flexibility required for next-generation AI infrastructure.

OCP SROI is Defining the System Requirements

- Scale-up fabrics create a different optical problem than traditional data-center networking: shorter reach, higher radix, tighter latency, and greater sensitivity to cost, power, and reliability.
- Optical connectivity must be evaluated at the system level, against the cost and performance benchmarks established by passive copper.
- OCP SROI (Short-Reach Optical Interconnect workstream) provides a shared requirements framework that can support multiple protocols, architectures, implementations, and module-placement strategies.
- As AI fabrics expand into multi-rack pods, optical interconnects are quickly becoming an architectural necessity rather than an incremental enhancement.



Takeaway
SROI is defining the system-level requirements that optical solutions must meet to scale within AI fabrics.

Multiple Levers Exist to Meet Scale-Up Requirements

- System designers have a range of architectural, electrical, and integration “levers” to optimize for their specific goals.
- Scale-up requirement categories: power, cost, reliability, latency, density, interoperability
- Some of these “levers” are shown below:

Copper / Optics Balance

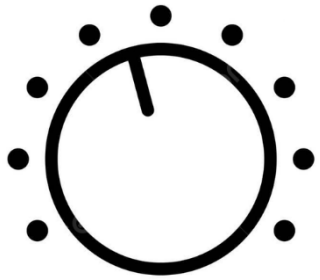
Active/ Retimed



Passive Copper — Linear Optics

Optical Placement

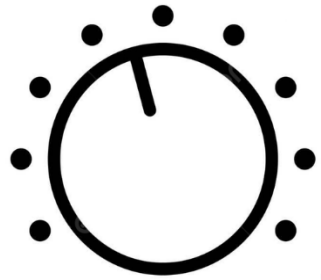
Near Package



Front Panel — On Package

System Integration

Modular



Discrete — Tight Integration

Industry Alignment Enables Faster Innovation and Deployment

- As multiple architectures and implementation approaches emerge, alignment helps reduce duplicated development effort and accelerate deployment.
- Industry efforts such as OCP SROI, OIF NPO project, CPX MSA, and related activities are helping establish common requirements, interfaces, and building blocks across emerging near-package architectures.
- Molex participates across these industry efforts and develops flexible interconnect solutions that support multiple near-package architectures within an increasingly interoperable ecosystem.

Takeaway
As AI scale-up architectures continue to accelerate, modularity and interoperability, established by industry alignment, will be increasingly required to enable the required pace of deployment.

Molex Solutions Across the Near-Package Ecosystem

SYSTEM REQUIREMENTS & METRICS

OCP SROI

Goal: Define what optical scale-up interconnects must achieve

OPEN INTERFACES & SPECIFICATIONS

OIF NPO • Open CPX • Related MSAs

Goal: Define interfaces for connectivity and interoperability

INDUSTRY ALIGNMENT

Shared system objectives, open interfaces, and interoperable building blocks

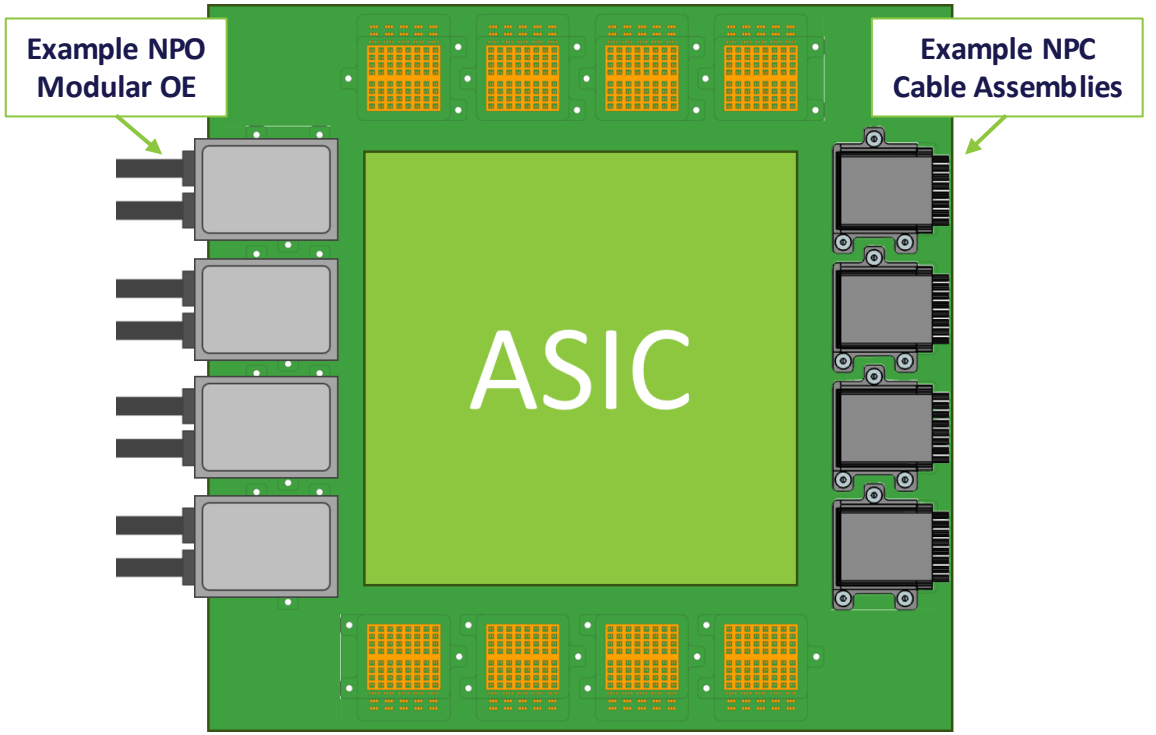
MULTIPLE NEAR-PACKAGE ARCHITECTURE PATHS

MOLEX FLEXIBLE INTERCONNECT PORTFOLIO

- **Mirror Mezz Enhanced**
- **Gemini Mezz**
- **CuOperate**
- **Si-Fly HD / CPX**

Interconnects Enable the Modular Near-Package Channel

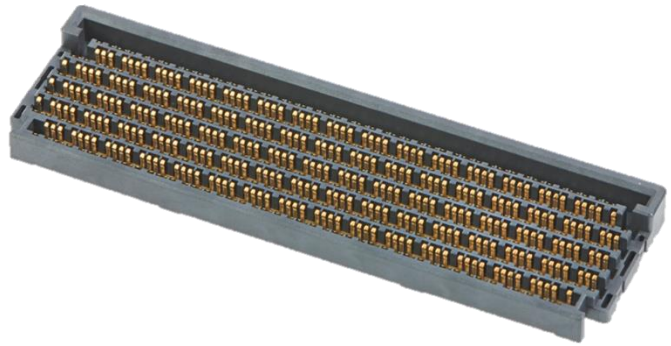
- Near-package architectures require a reliable, manufacturable transition between the PCB and the optical or copper endpoint.
- NPO Mezzanine: Board-to-Module Interface
  - Connects the host PCB to removable optical engine (OE)
  - Maintains short electrical path close to the ASIC
  - Supports high-speed signaling, power, control, alignment, and module retention
  - Enables modular assembly, replacement, and OE choice
- NPC Cable Assembly: Board-to-Cable Interface
  - Transitions the host PCB signals directly into twinax
  - Extends copper beyond practical routing distance thru PCB
  - Provides positive retention, strain relief, and cable exit
  - Routes the electrical channel to a panel-located endpoint



Takeaway
Modularity and flexible interconnect remain critical as copper and optics continue to coexist in the near-package environment supporting scale-up links

Mirror Mezz Enhanced: A Proven Mezzanine Option for Modular NPO

- A genderless (self-mating) 224G capable mezzanine solution, Mirror Mezz Enhanced enables modular, pluggable NPO
- Low-profile stack height of 5mm with integrated gatherability and mechanical wipe to support challenging stackups
- Scalable design supports multiple configurations for various module sizes through insert-molded terminal architecture
- Mirror Mezz Enhanced builds on the Mirror Mezz family, which is incorporated into OCP OAI/OAM v1.0 and v1.5 specifications and has been deployed in volume for several years

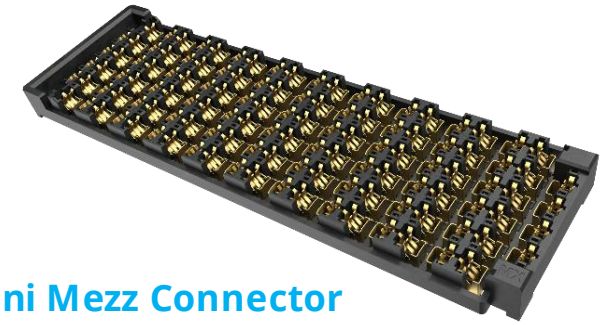




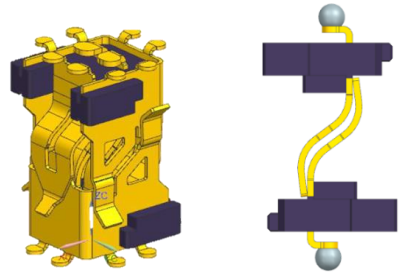
Takeaway
Mirror Mezz Enhanced extends a proven OCP mezzanine architecture into 224G modular NPO applications, providing an available, low-risk path to next-generation connectivity needs.

Gemini Mezz: An Optimized Interface For 6.4T and 7.2T NPO Modules

- Designed for 224G NPO Systems: New, individually-shielded differential pairs improve 224G channel performance
- Gemini Mezz preserves proven mezzanine benefits from Mirror Mezz:
  - Genderless, self-mating PNs
  - 5mm stack height
  - BGA termination to PCB
  - Protected pin geometry
  - Scalable design
- System Integration: A symmetric 3mm x 3mm pair pitch simplifies and improves PCB breakout and routing
- Ecosystem Alignment: Molex has expedited the tooling and release of an 8P x 13R version of Gemini Mezz
  - Single interface for 6.4T and 7.2T optical modules
  - Foundation for a common NPO pin-map strategy
  - Proposed pin map details are shown on the following slide



Gemini Mezz Connector

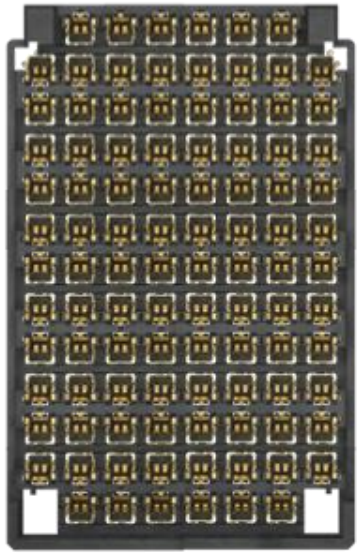


Fully Shielded Differential Terminals

Takeaway
Gemini Mezz combines improved 224G performance with a common interface capable of supporting multiple NPO module classes

Gemini Mezz: An Optimized Interface For 6.4T and 7.2T NPO Modules

One 100DP interface spans 6.4T & 7.2T



GEMINI MEZZ · HERMAPHRODITIC INTERFACE

8x13	100DP	224G
PIN MATRIX	TOTAL CAPACITY	DATA RATE

100 differential pairs, strategically allocated

A common interface supports both module bandwidth classes by unifying high-speed I/O, control, and 36A power in one pluggable mezzanine footprint.

	SE	12V	12V	12V	12V	SE	
Rx	Rx	SE	12V	12V	SE	Rx	Rx
Rx	Rx	SE	12V	12V	SE	Rx	Rx
Rx	Rx	Rx	SE	SE	Rx	Rx	Rx
Rx	Rx	Rx	Rx	Rx	Rx	Rx	Rx
Rx	Rx	Rx	(Rx)	(Rx)	Rx	Rx	Rx
SE	Rx	Rx	(Rx)	(Rx)	Rx	Rx	SE
Tx	Tx	SE	3V	3V	SE	Tx	Tx
Tx	Tx	SE	1.8V	1.8V	SE	Tx	Tx
Tx	Tx	Tx	SE	SE	Tx	Tx	Tx
Tx	Tx	Tx	Tx	Tx	Tx	Tx	Tx
Tx	Tx	Tx	(Tx)	(Tx)	Tx	Tx	Tx
	Tx	Tx	(Tx)	(Tx)	Tx	Tx	

8 PAIRS × 13 ROWS

64DP — **6.4T payload**
32 Tx + 32 Rx

+8DP — **7.2T extension**
4 Tx + 4 Rx

16DP — **Control**
32 single-ended pins

12DP — **Power**
36A delivery

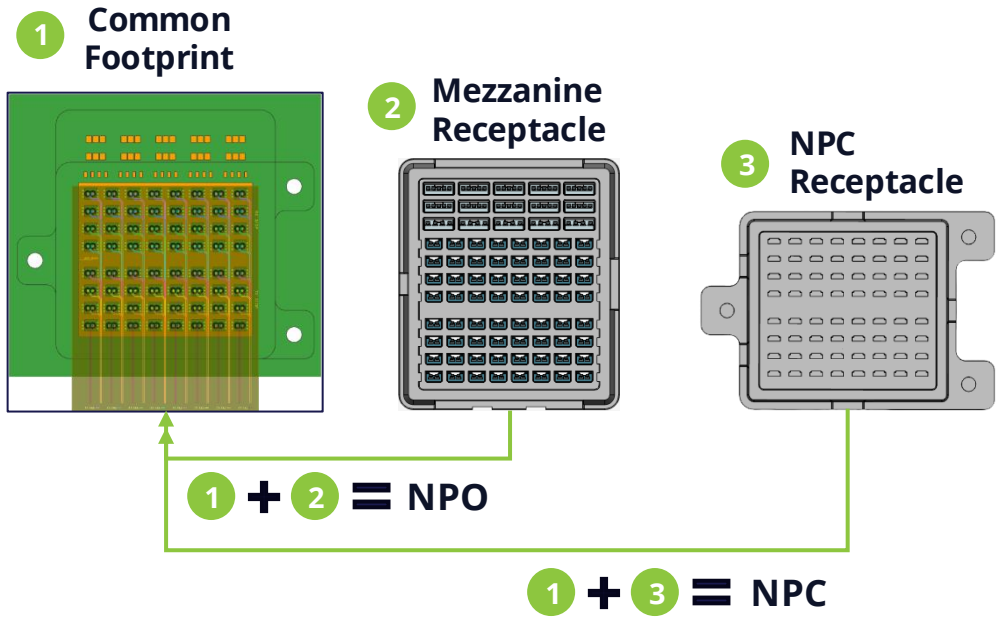
TOTAL CAPACITY



64 + 8 + 16 + 12 = 100DP

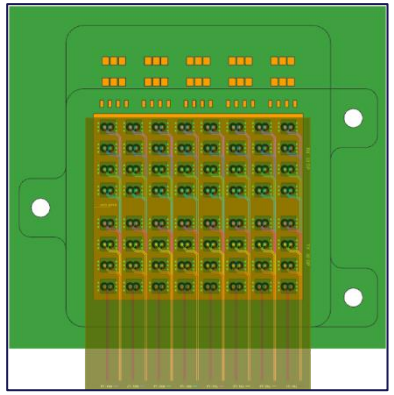
CuOperate: One Product Supporting Both NPO and NPC

- Traditional near-package architectures typically require separate footprint definitions for NPO mezzanine modules and NPC cable implementations.
- CuOperate is a **new product** from Molex that introduces a common host board footprint supporting both NPO and NPC through purpose-built connector solutions: CuOperate Mezz and CuOperate NPC.
- The shared platform preserves architecture choice while allowing each interface to remain optimized for its intended application.
- System architects have more flexibility with CuOperate to adjust their “levers” according to the different requirements discussed previously

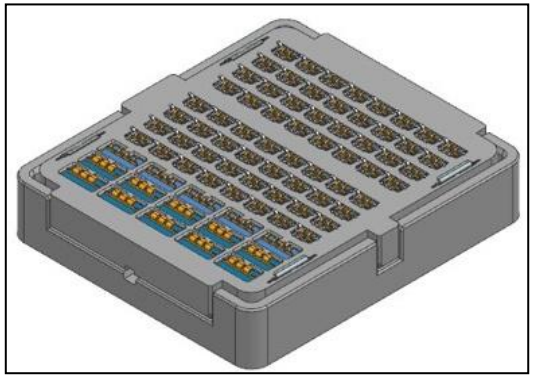


Takeaway
CuOperate separates platform decisions from architecture decisions through a common host-side footprint supporting both NPO and NPC implementations.

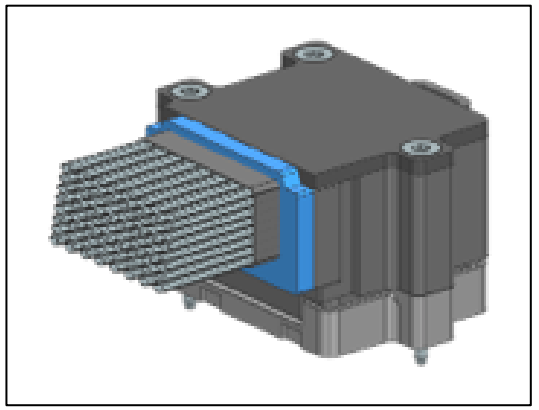
CuOperate: One Product Supporting Both NPO and NPC



Common host-board elements
High-speed land pattern
Differential-pair arrangement
Grounding structure
Board escape strategy
Mechanical datum strategy

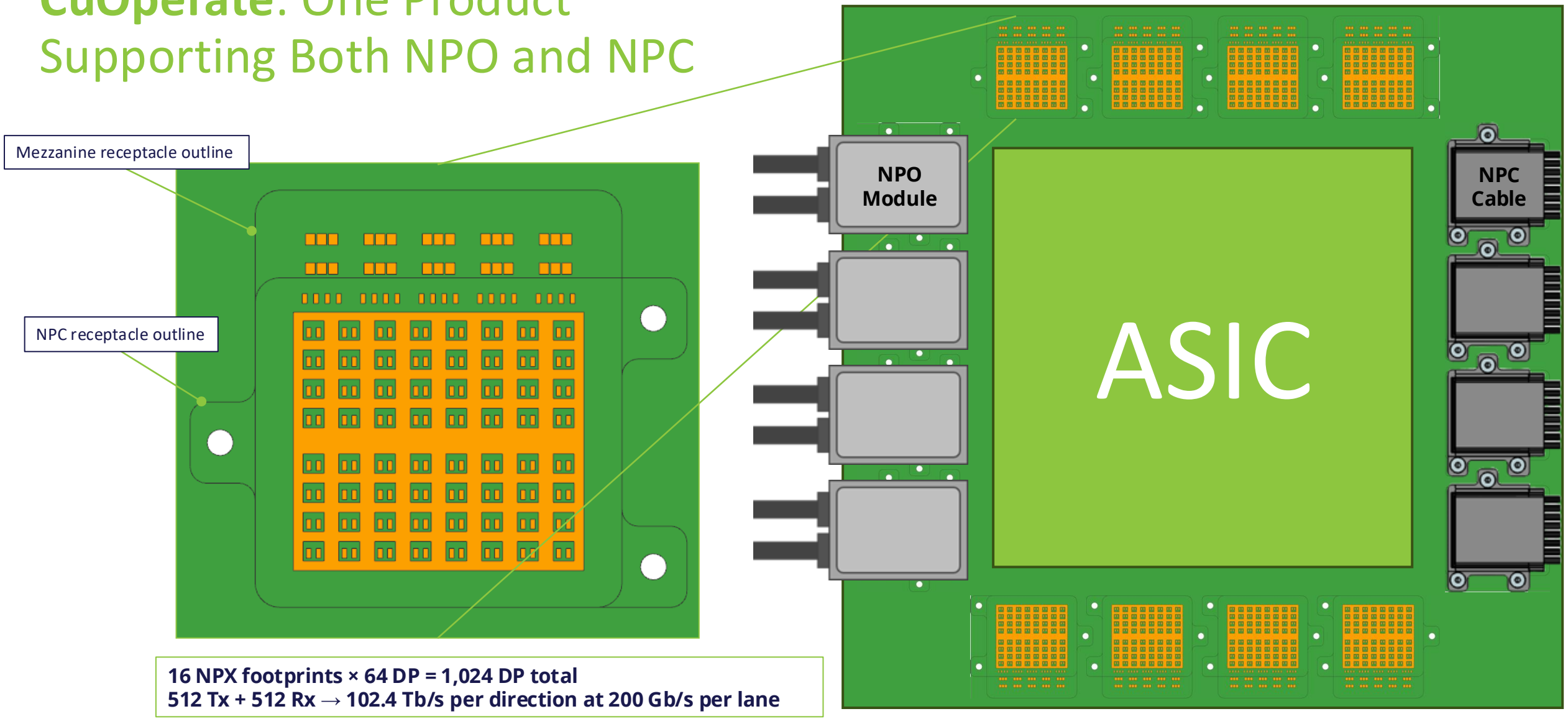


NPO-specific elements
Mating mezzanine connector
Power contacts
Sideband contacts
Module guidance
No positive retention required



NPC-specific elements
Cable plug and twinax
Right-angle cable exit
Screw retention
Cable strain management
Cable mating ergonomics

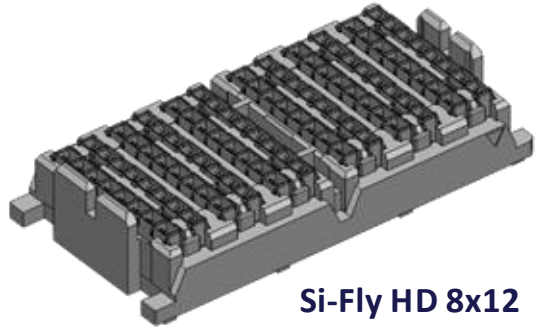
CuOperate: One Product Supporting Both NPO and NPC



16 NPX footprints × 64 DP = 1,024 DP total
512 Tx + 512 Rx → 102.4 Tb/s per direction at 200 Gb/s per lane

Si-Fly HD: Enabling the Open CPX Ecosystem

- Si-Fly HD is a Molex-licensed, interoperable connector system supporting the Open CPX ecosystem today
- Molex is a founding member of the Open CPX MSA and is actively involved in development, contribution, and implementation
- The MSA enables support for interoperable building blocks across suppliers through second-sourcing agreements and tooling commitments



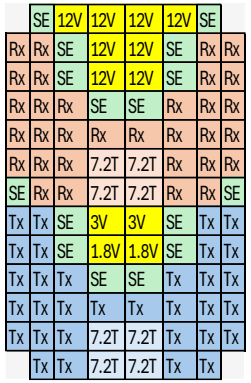
Si-Fly HD 8x12
CPX Type 2

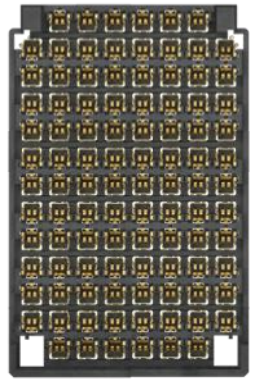
OPEN CPX MSA

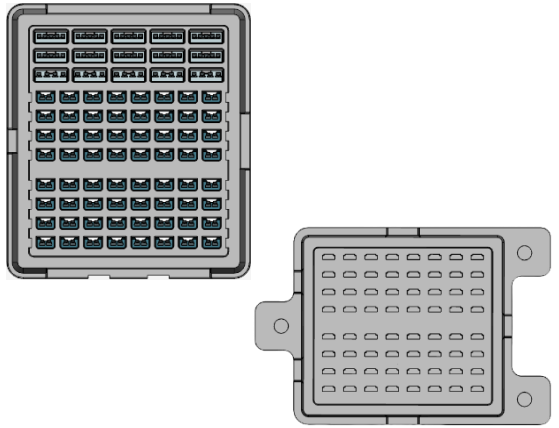
Takeaway
Si-Fly HD demonstrates Molex's commitment to enabling open hardware ecosystems through industry partnerships and multi-source agreement activities

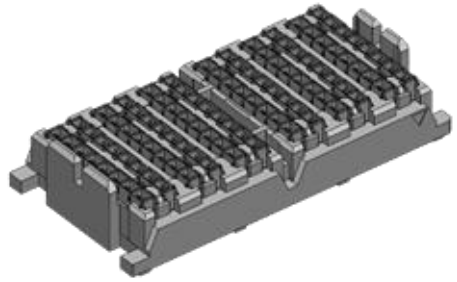
Molex Interconnect Solutions to Support Near-Package AI Scale-Up











Mirror Mezz Enhanced

- Proven modular solution
- OCP heritage
- Production-validated architecture at 224G

Gemini Mezz

- Low-profile, high-performance
- 8x13 version for 6.4T and 7.2T support
- Proposed common reference pin map

CuOperate

- Flexible NPO / NPC platform
- Shared PCB footprint supporting both NPO and NPC implementations
- Architecture choice preserved

Si-Fly HD

- CPX implementation
- Multi-source support
- Open ecosystem enablement

No single interconnect architecture will satisfy every AI scale-up application. Molex is investing across multiple solutions to give system architects greater flexibility as optical and copper ecosystems continue to evolve and solidify.

Let's Build Together – Call to Action

Engage with Molex



Full-channel SI modeling

Co-develop end-to-end channel simulation with our models – mezzanine and cable – against your ASIC, escape and reach budget.



Cost & system impact study

Quantify the alternatives together: SKU reduction, optics vs. copper cost, mechanical implications, power at the platform level.



Footprint review

Evaluate footprint options against your ASIC escape and mechanical datum strategy before it's frozen.



Samples & design-in support

Functional samples for measurement & evaluation. System application guides and mockup resources available.

Visit Molex Booth D21 to learn about these solutions and more!

molex

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Thank You

Questions?

